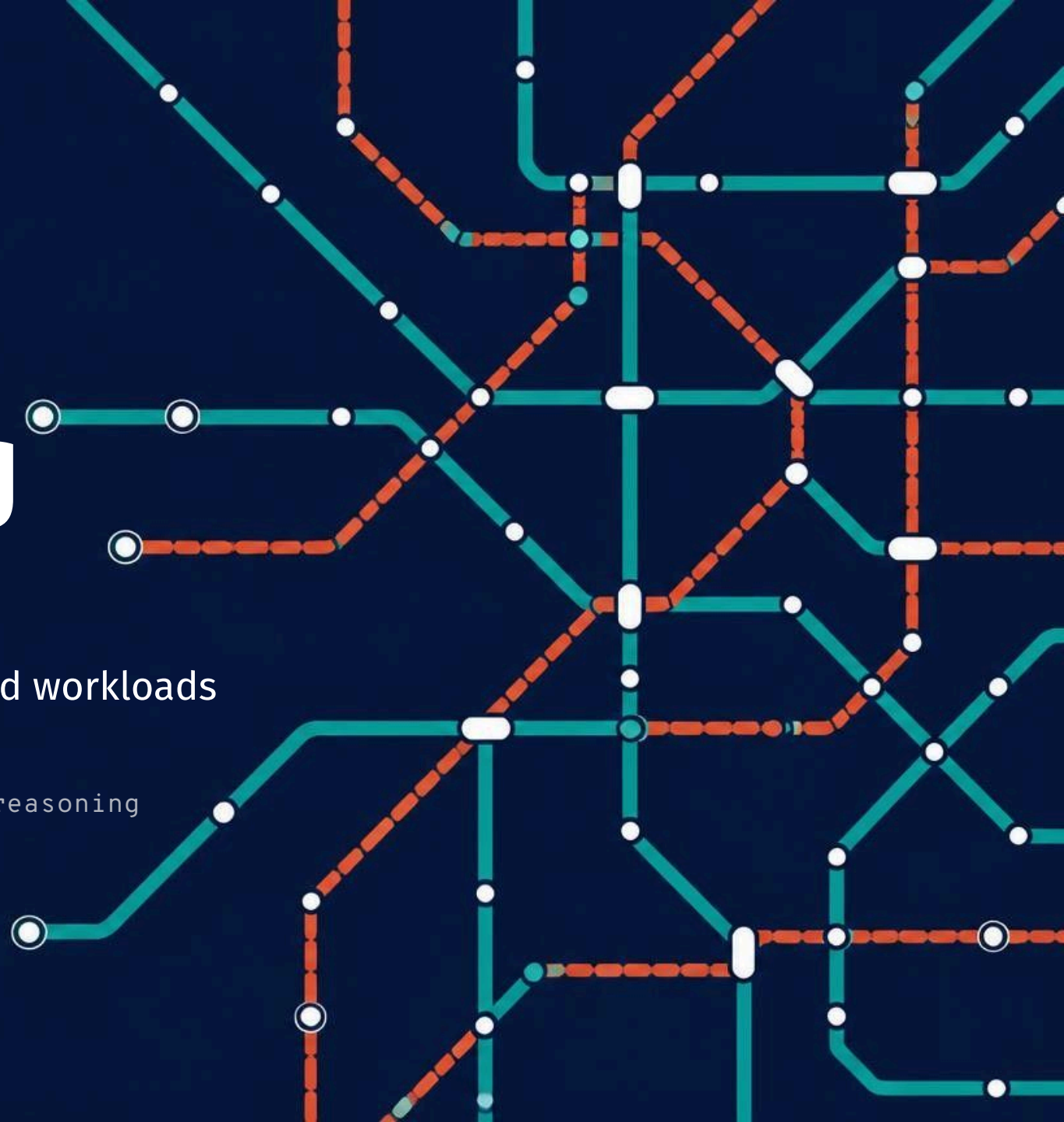
CYBERSECURITY DEEP DIVE

Cloud Computing Security

How identities, configurations, interfaces, and workloads become attack paths

University security foundations · threat-first reasoning



01 · THE SHIFT

Why the cloud changes the security problem

The platform is only one part of the system.



Programmable

Resources can be created, changed, and deleted through code or APIs.



Shared and reachable

Services are pooled, internet-reachable, and managed across organizational boundaries.



Gap-prone

Failures often come from identity, configuration, or responsibility gaps.



02 · THE FOUNDATION

Cloud computing in one mental model

On-demand access to shared computing resources delivered over a network.

Elasticity

Scale capacity up or down as demand changes.

Measured use

Consumption is visible and attributable.

Pooled resources

Many customers draw from shared infrastructure.

Self-service

Users provision capability without waiting for a central operator.



03 · CONTROL BOUNDARIES

IaaS, PaaS, and SaaS shift the control boundary

MODEL	PROVIDER MANAGES	CUSTOMER STILL OWNS
IaaS	Physical infrastructure and core virtualization.	Operating systems, workloads, identities, and data.
PaaS	More of the platform and runtime.	Code, configuration, identities, and data.
SaaS	Application and underlying platform.	Users, access, data, and configuration.

04 · DEPLOYMENT MODELS

Public, private, hybrid, and multi-cloud

The model changes the attack surface, trust boundaries, visibility, and operational complexity.

Public

Provider-operated infrastructure shared across customers.

Private

Dedicated environment with tighter organizational control.

Hybrid

Connected environments create integration paths and policy seams.

Multi-cloud

Multiple providers increase duplicated identities and inconsistent enforcement.



05 · OWNERSHIP

The shared responsibility model

Security duties are divided between the cloud service provider and cloud customer. The boundary is different for every service.

PROVIDER

Secure the cloud

Facilities, hardware, core services, and the platform commitments defined by the service.

CUSTOMER

Secure what runs in it

Each workload, identity, data store, configuration, and network path.

CSA guidance emphasizes aligning controls with cloud-native architecture and the shared responsibility model.

Threats are paths to impact

1. **Weakness**
   Identity, configuration, code, interface, data, or process.
2. **Exploit**
   A threat actor or accident crosses a control boundary.
3. **Access**
   Resources, permissions, secrets, or control-plane actions become available.
4. **Impact**
   Exposure, disruption, fraud, lateral movement, or loss of control.

The useful question is not “is the cloud secure?” It is “what path reaches what impact?”

Identity is the new perimeter

Compromised credentials, excessive permissions, stale accounts, and weak federation create high-leverage access. A valid identity can travel through the cloud faster than a network boundary can stop it.



Reduce entry

Phishing-resistant authentication, federation hygiene, and short-lived access.



Watch use

Detect unusual identity activity before it becomes persistence.



Limit reach

Least privilege, scoped roles, and removal of stale accounts.

Misconfiguration: the most preventable cloud failure



Public storage

A private data store becomes reachable from the internet.



Disabled logging

A preventable error becomes invisible.



Permissive security groups

Exposed management ports and broad rules open control paths.



Prevent drift

Default settings can be reduced with secure defaults, policy-as-code, continuous scanning, and change review.

Insecure APIs and control-plane abuse

Cloud APIs expose powerful operations for provisioning, data access, and privilege changes.

Weak authorization, poor input validation, leaked keys, and missing rate limits turn APIs into attack paths.



Defensive checks: strong authorization · input validation · key hygiene · rate limits · audit trails

Data exposure and weak cryptographic controls

Risk comes from exposed databases, unmanaged backups, overbroad sharing, and secrets in code or logs.

1. **Classify**

   Identify sensitive records, regulated data, backups, and secrets.

2. **Protect**

   Encrypt in transit and at rest; manage keys deliberately.

3. **Constrain**

   Control sharing, snapshots, database access, and service identities.

4. **Monitor**

   Detect unusual reads, exports, and access from unexpected paths.

Supply chain and workload threats

Every dependency adds another path worth defending.



Images, packages, managed services, and third-party integrations extend trust.

CI/CD pipelines extend trust.

Attackers can poison dependencies, implant code, steal build credentials, or pivot through a vulnerable workload.



Credential theft to cloud takeover

1. **Phish or steal**

   Capture a user credential or access token.

2. **Reuse identity**

   Present it to a cloud console or API.

3. **Discover**

   Enumerate resources, roles, and reachable permissions.

4. **Take over**

   Escalate privilege, access data, or create persistence.

Break the path with phishing-resistant authentication, least privilege, short-lived access, and identity telemetry.

Exposed storage to data breach

1. **Find exposure**
   Locate a public bucket, share, snapshot, or backup.
2. **Enumerate contents**
   Identify sensitive records and useful metadata.
3. **Exfiltrate or reuse**
   Remove data or use it for follow-on attacks.
4. **Stay hidden**
   Exploit weak monitoring or unclear ownership.

Exposure is often a configuration problem before it is a cryptography problem.

Vulnerable workload to lateral movement

1. **Exploit workload**

   Compromise an internet-facing application or container.

2. **Harvest credentials**

   Read metadata credentials or service tokens.

3. **Move laterally**

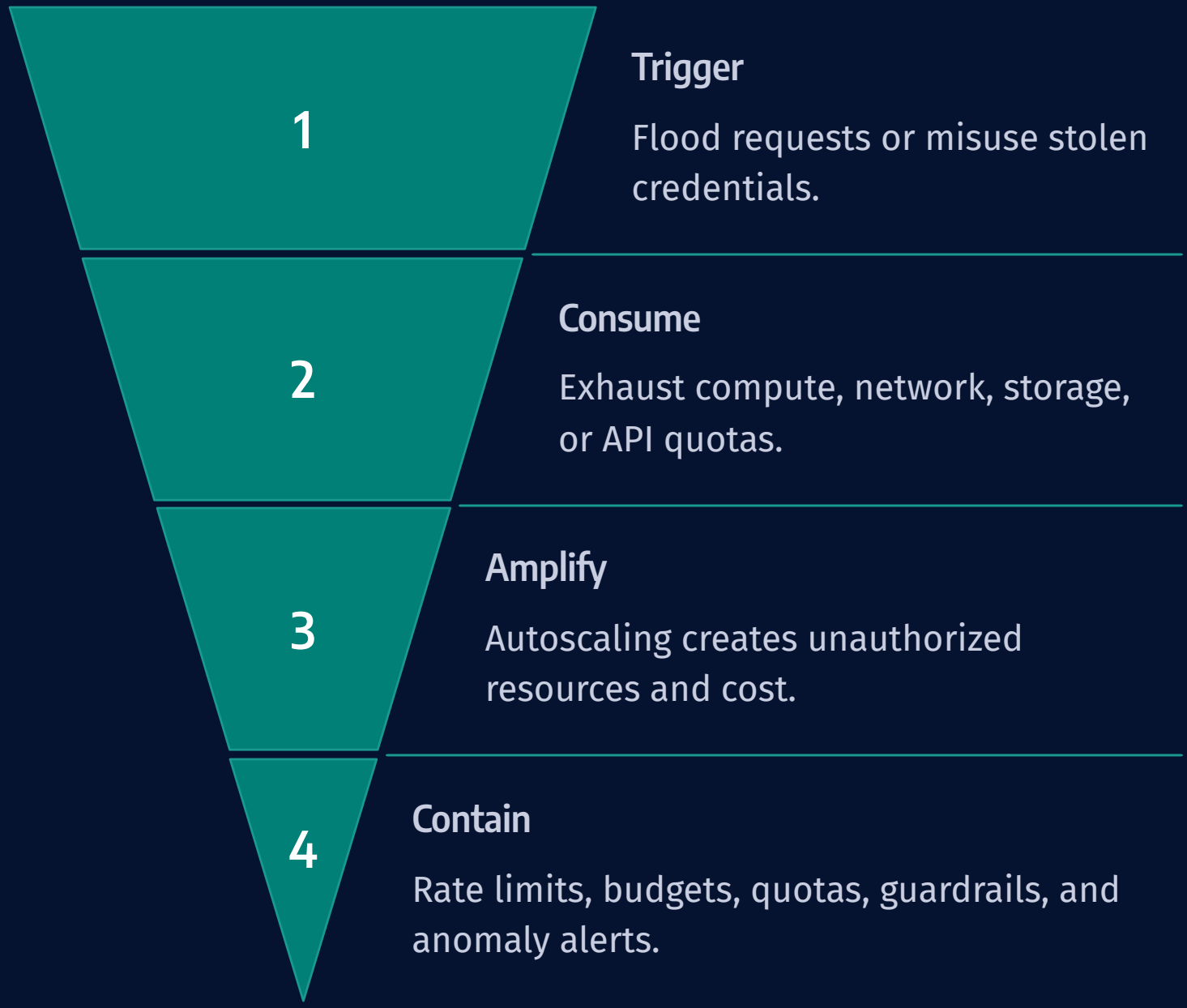
   Use flat networks and overprivileged roles.

4. **Impact neighbors**

   Reach adjacent workloads, data, or control-plane services.

Denial of service and resource abuse

Cloud elasticity helps legitimate demand, but it can also amplify abuse.



Security architecture: defense in depth

Identity	Strong authentication, least privilege, short-lived access.
Network	Segmentation, private paths, and constrained egress.
Workload	Harden hosts, images, containers, and runtime behavior.
Data	Classify, encrypt, manage keys, and monitor access.
Detection and recovery	Centralize telemetry, contain compromise, and rehearse restoration.

Assume one control will fail. Design the next layer to detect or contain the breach.

Cloud security operating model

Govern

Inventory assets, classify data, define ownership, and set policy.



Protect

Apply least privilege, secure configuration, encryption, segmentation, and secure delivery.

Detect

Centralize telemetry and investigate identity activity.

Respond

Contain access and rehearse recovery before the incident.

Governance becomes useful when it changes daily engineering behavior.

Cloud security tools by job to be done

Find drift

CSPM and policy-as-code tools find misconfiguration and drift.

Protect workloads

CWPP, container, Kubernetes, and vulnerability scanners protect runtime assets.

Control identity

IAM analyzers, privileged access management, MFA, and secrets managers reduce access risk.

Connect signals

SIEM, cloud-native detection, SOAR, and network analytics connect evidence to response.

Choose the control objective first, then the product category.

Frameworks make controls measurable

“Secure the cloud” becomes actionable when every control has an owner, evidence, and a due date.

CONTROL ELEMENT	QUESTION TO ANSWER	EVIDENCE
Asset	What service, workload, identity, or data store is in scope?	Inventory record
Owner	Who is accountable for the control?	Named person or team
Test	How do we know the control works?	Continuous or periodic check
Exception	What is accepted, until when, and why?	Risk decision
Remediation	What changes next, and by what date?	Tracked action

CSA’s Cloud Controls Matrix is a vendor-neutral framework for cloud security and privacy controls; its current introduction describes 207 controls across 17 domains.

Start with the path, not the product.

A strong cloud security assessment asks where an attacker can enter, what they can reach, and what proves the controls work.

1. Start with identities, internet exposure, data stores, privileged paths, and logging coverage.
2. Trace one realistic attack path from initial access to business impact.
3. Make every control testable, continuously monitored, and owned by a person or team.

